LAND MOBILE SATELLITE COMMUNICATIONS: A FURTHER DEVELOPMENT IN INTERNATIONAL SPACE LAW (PART I)

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Introduction

It is now approximately ten years since the International Maritime Satellite Organization (INMARSAT) came into existence.[1] During that period it has experienced rapid growth in its membership, its user community and its revenues. At present there are fifty-five Member States. Approximately eight thousand ships are fitted with earth stations communicating via the INMARSAT system. In 1988 the total revenues of the Organization were US $98.8 million, an increase of 34 per cent over the preceding year.

There has also been an increase in the range of services provided through the INMARSAT space segment. For example, while telephone, telex, facsimile and data services have long been offered to ships, there are now the additional possibilities of compressed video and slow-scan television. The organization is also developing navigation and radiodetermination services.

The most striking transformation, however, in the nature of the Organization is without doubt the widening of its institutional competence, which will enable it to serve entirely new user groups. The Organization was originally established "...to make provision for the space segment necessary for improving maritime communications, thereby assisting in improving distress and safety of life at sea communications, efficiency and management of ships, maritime public correspondence

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The views expressed in this article are those of the authors and are not necessarily those of any organization with which the authors are or have been connected.

1. *See* H. H. M. Sondaal, *The Current Situation in the Field of Maritime Communications Satellites: "INMARSAT"* , 8 J. SPACE L. 9 at 34 (1980).

services and radiodetermination capabilities."[2] In 1985 the INMARSAT Assembly adopted amendments to the INMARSAT Convention and Operating Agreement which gave the Organization the additional competence to provide aeronautical satellite telecommunications. The history and nature of these amendments have already been described in the pages of this Journal.[3] It is now expected that the aeronautical amendments will come into force at some time in 1989.[4] Meanwhile, commercial trials of aeronautical satellite communications using the INMARSAT space segment are already in progress;[5] and certain operational services have been offered on an Interim basis, for example, to the Ontario air ambulance service.[6]

In January 1989, an extraordinary session of the INMARSAT Assembly adopted further amendments to the Convention and Operating Agreement. These confer on the Organization the competence to provide land mobile-satellite communications. The amendments will enable INMARSAT to respond both to existing and to predicted future demands for land mobile communications - maritime, aeronautical and land mobile - will have considerable operational and economic advantages.

In the first part of this article it is intended to describe briefly the potential applications of land mobile satellite communications, and the limited services of this type which have already been provided through the INMARSAT space segment. An explanation will be given of the institutional basis on which these limited services have been offered. We will then describe the history and origins of the recent amendments to the INMARSAT constituent instruments. This will involve consideration of the changes to the Radio Regulations which were agreed at WARC MOB-87. In the second part of this article[7], the amendments will be analyzed in detail. We shall conclude with an overview of the competitive and regulatory framework within which international land mobile-satellite services will be provided.

Applications

It is helpful to introduce at this point certain definitions. The

2. Convention on the International Maritime Satellite Organization (INMARSAT), with annex, Sept. 3, 1976, Art. 3 (1), 31 U.S.T. 1, T.I.A.S. No. 9605. This Convention will be referred to in subsequent footnotes as "CONV".
3. *See* W. D. von Noorden, *Space Communications to Aircraft: A New Development in International Space Law*, 15 J. SPACE L. 25, 147 (1987).
4. For an explanation of the amendment process, see *id.* 148-9.
5. AERONAUTICAL SATELLITE NEWS, December 1988, at 1.
6. COUNCIL/27/SR/FINAL, 13.3.4.
7. The second part of this Article will appear in a future issue of the Journal.

Radio Regulations define "land mobile earth station" as "a mobile earth station in the land mobile-satellite service capable of surface movement within the geographical limits of a country or continent."[8] They also define "mobile earth station" as "an earth station in the mobile-satellite service intended to be used while in motion or during halts at unspecified points."[9] These definitions obviously include earth stations which are fitted to vehicles, such as trucks and trains, and which may be used whether the vehicle is in motion or stationary. The definitions also include earth stations which are light and compact enough to be carried readily from place to place, but which are always stationary when in use. It may be noted that the definitions would further apply to a hand-held "global personal communicator"; but that is to anticipate future technology.[10]

The potential users of land mobile-satellite communications are very diverse, consisting for example of trucking companies, car hire firms, container shippers and railway organizations.[11] Considerable use is already made of transportable earth stations in remote regions where no alternative telecommunications facilities exist; and in areas where the existing telecommunications infrastructure has been disrupted in the aftermath of natural disasters.[12]

The land mobile market is known to be highly differentiated, encompassing both basic and sophisticated user requirements. The range of required services is currently seen to be: two-way messaging; mobile-to-land position reporting; mobile-to-land data reporting; land-to-mobile polling; paging; land-to-mobile group calls; emergency alerting; voice services; and radiodetermination.[13]

It is believed that the overall market for land mobile satellite communications will be only a small percentage - perhaps 2 to 5 percent- of the total land mobile communications marketplace, which will continue to be satisfied largely by conventional (terrestrial) land mobile radio systems, including cellular. However, in some less populous regions of the world the cost of a cellular infrastructure cannot be justified. It is acknowledged that satellites have a role to play in these circumstances.[14]

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8. Art. 1, Sec. 4.12A, as inserted by WARC MOB-87.
9. Art. 1, Sec. 4.9.
10. "Satellites and Their Role In The Mobile Revolution," a speech by Olof Lundberg to the *Financial Times* Conference on the Outlook for World Mobile Communications, London, 8 November 1988.
11. ASSEMBLY/6/2, ATTACHMENT, sec. 2.1.
12. INMARSAT ANN. REV. 1987-88, at 18.
13. *See supra* note 11.
14. *Id.*

Even in areas served by cellular systems, mobile satellite services offer a superior data communications facility.[15]

The market for land mobile-satellite communications also includes those such as the operators of long-distance trucks, whose journeys would normally cross many regional and national cellular boundaries and who would prefer therefore to fit a single piece of equipment, as would be possible with a global satellite system, in preference to multiple cellular units. Even this limited part of the overall land mobile communications market represents a potential for terminal fittings far in excess of those previously anticipated for INMARSAT in its maritime and aeronautical sectors.[16]

Existing INMARSAT Practice

Although INMARSAT had originally the express competence to provide only maritime communications, the Organization has permitted land-based earth stations to use its space segment on an exceptional basis. In particular, the Council has authorized the Director General to grant access to the INMARSAT space segment for emergency relief operations on land.[17] This has been done in the aftermath of natural disasters such as earthquakes, floods, hurricanes and volcanic eruptions, when terrestrial communications have been disrupted.[18] The Council has also authorized the Director General to grant, on special conditions, access to the space segment for more general land mobile applications.[19] Under this authority access has been granted, for example, to companies exploring for minerals in remote areas and to a mountain climbing expedition.[20]

The justification for permitting such uses may be found in particular in Article 5 (3) of the INMARSAT Convention, which requires the Organization to operate "on a sound economic and financial basis having regard to accepted commercial principles". It is a generally accepted commercial principle that an organization should derive whatever revenue it can from surplus assets. In the case of INMARSAT the surplus asset is residual space segment capacity, which may be made

15. Rober T. Gallagher, *Land-Based Satellite Services for Mobile Communications*, TELECOMMUNICATIONS 59 (Nov. 1988).
16. *See supra* note 11.
17. COUNCIL/11/SR/FINAL, 16.31.
18. "The Virtue of Impatience", an address by Olof Lundberg to the IAF/COSPAR Symposium on Space Communications for Development, New York, 18 February 1987.
19. COUNCIL/21/SR/FINAL, 12.2.8. Previously the Council had authorized such access on a case-by-case basis.
20. *See supra* note 12.

available for non-maritime communications.[21] Since this helps to meet the overall costs of the space segment, it permits the charges for maritime communications to be lower than would otherwise be the case, and therefore indirectly serves the express purposes of the Organization.

This rationale for permitting land-based communications through the INMARSAT space segment has consequences for the terms and conditions which the Council imposes in such cases. Since only residual space segment capacity may be used, land mobile communications are subject at all times to the prior and overriding requirements of INMARSAT's maritime communications services; and therefore must be subject to preemption in favour of maritime services.[22] There are also consequences for space segment planning. Whatever the demand for land-based communications, the Organization has not previously been entitled to take this into account in determining the size of its future space segments.

It should be mentioned that the INMARSAT Council has also authorized the use of the space segment for land-based fixed communications. The conditions referred to above apply equally in such cases; and it is an additional requirement that there should be no reasonable alternative telecommunications facilities available.[23] INMARSAT recognizes that fixed-satellite communications are within the express competence of the INTELSAT and certain regional satellite organizations. In the past it has not always been predictable, particularly in remote regions, to install earth stations communicating through the INTELSAT space segment. This situation is changing with the introduction of very small aperture terminals (VSATs) which may reduce the demand for fixed-satellite communications through INMARSAT.

Land-based earth stations, whether fixed or mobile, already represent a significant percentage of the total earth stations commissioned within the INMARSAT system.[24] They all, however, conform to the INMARSAT standard known as Standard-A, which was intended for maritime applications. The potential mass market for land mobile-satellite telecommunications will be addressed by INMARSAT with new technologies. It will also be addressed on a sound institutional basis, the origins of which will now be considered.

History and Origins of the Land Mobile Amendments

In its report to the Fifth Session of the INMARSAT Assembly, which was held in October 1987, the INMARSAT Council referred to land-

21. In the past this has also been a basis for authorizing the use of the space segment for aeronautical satellite communications: *e.g.*, COUNCIL/18/SR/FINAL, 16.3.4.
22. COUNCIL/21/SR/FINAL, ANNEX IX, para. (1).
23. COUNCIL/18/SR/FINAL, 12.5.4(a) and ANNEX VII, para. (g).
24. *See infra* note 25.

based non-maritime services.[25] It pointed out that, as of 31 August 1987, a total of 154 land-based ship earth stations were commissioned to utilize the space segment for non-maritime communications. INMARSAT had developed the Standard-C ship earth station as a very small, low cost ship earth station for exchanging messages and data.[26] The Standard-C system had been defined and specified for the marine environment, but was under evaluation for land mobile-satellite applications. "In this regard", the Council reported, "the Director General is assisting a group of European Signatories in the preparation of a trials programme, to investigate the technical suitability and potential market acceptance for such a land mobile-satellite services, and the fact that INMARSAT has obtained wide experience and facilities in the provision of mobile-satellite services, the Council recognized that the Assembly might wish to consider the enhancement of INMARSAT's institutional competence to this effect."[27] It should be noted, however, that the Assembly does not have the power to initiate amendments to the INMARSAT Convention and Operating Agreement.

One of the functions of the INMARSAT Assembly is to "consider and review the activities, purposes, general policy and long-term objectives of the Organization and express views and make recommendations thereon to the Council.[28] In this context the Assembly considered the report of the Council and responded with two crucial decisions:

"(a) to recommend that the council examine the commercial, technical and operational feasibility of providing land mobile-satellite services, taking into account the different interests of Parties; and

"(b) to draw the attention of Parties and Signatories to the discussions which have taken place at the Fifth Session of the Assembly on this subject and to the procedures for initiating amendments to the INMARSAT to provide land mobile-satellite services, in accordance with Article 34(1) of the Convention and Article XVIII(1) of the Operating Agreement."[29]

The reference, in the first of these decisions, to "the different interests of Parties" is partly explained by a statement by the delegations of Canada, India and the USA which was annexed to the Report of the Assembly: "The Representatives of Canada, India, and the United States of America noted that certain INMARSAT Parties are planning domestic

25. ASSEMBLY/5/1, 10.4.1.
26. *Id. at* 10.4.2.
27. *Id. at* 10.4.2. and 10.4.3.
28. CONV art. 12(1)(a).
29. ASSEMBLY/5/11, 5.9.3.

satellite systems to provide mobile services and requested that the Council recognize this in its feasibility studies.[30]

After the Fifth Session of the Assembly events moved with remarkable speed. Little more than a month later, on 30 November 1987, the Federal Republic of Germany formally initiated the procedures for amendment of the Convention and Operating Agreement by submitting proposed amendments to the Director General, together with explanatory comments. The amendments will be considered in detail in the second part of this article, but their essence may be found in two provisions. It was proposed that a paragraph should be added to the Preamble to the Convention, "Affirming that a maritime and aeronautical satellite system shall also be open for land mobile communications and communications on waters not part of the maritime environment for the benefit of all nations".[31] In addition, it was proposed that Article 3(1) of the Convention should be amended so as to read as follows:

> "The purpose of the Organization is to make provision for the space segment necessary for improving maritime communications and, as practicable, aeronautical and land mobile communications and communications on waters not part of the marine environment, thereby assisting in improving communications for distress and safety of life, communications for air traffic services, the efficiency and management of ships, aircraft and land transport, maritime, aeronautical and other mobile public correspondence services and radiodetermination capabilities."[32]

The Director General immediately circulated the proposed amendments to all INMARSAT Parties and Signatories for comment, as required by the INMARSAT constituent instruments.[33] At this time, the Twenty-Eighth Session of the Council was in progress. The Council noted the decisions of the Assembly and the initiative of the Federal Republic of Germany.[34] The Council also noted that the Director General was already carrying out a study of the commercial, technical, and operational feasibility of providing land mobile-satellite services which would address "the different approaches required for land mobile-satellite services when compared to the maritime and aeronautical services."[35]

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30. *Id. at* Annex VIII.
31. COUNCIL/29/13/REV/1, ANNEX II, page 8.
32. *Id.* at 19.
33. CONV art. 34(1); Operating Agreement on the International Maritime Satellite Organization (INMARSAT), Art XVIII(1). The Operating Agreement will be referred to in subsequent footnotes as "OA". *See also supra* note 4.
34. COUNCIL/28/SR/FINAL, 15.1.
35. *Id. at* 15.1.3.

The 1987 WARC

The Director General also reported to the Twenty-Eighth Session of Council on the results of the World Administrative Radio Conference for Mobile Services (WARC MOB-87) which was held in Geneva from 14 September to 17 October 1987.[36] INMARSAT Directorate staff members attended throughout the Conference as observers and participated in many of the Working Groups.[37] The Conference decided upon revisions to the Radio Regulations, certain of which were of great importance for the future of land mobile-satellite communications.

Prior to WARC MOB-87 there were frequency allocations for the maritime mobile-satellite service and the aeronautical mobile-satellite service, but none expressly for the land mobile-satellite service. The Conference decided to allocate 3 MHz in each direction within the existing maritime allocations for use by both land and maritime mobile-satellite services.[38] These bands were included in the specification of INMARSAT's Second Generation Satellites, which were then already under construction.[39] The remainder of the existing maritime allocation was opened up for the land mobile-satellite service, but on a secondary basis. A further 4 MHz in each direction was allocated adjacent to the aeronautical band for use by the land mobile-satellite service on a primary basis.[40]

The results of WARC MOB-87 represent a compromise between those who wanted more spectrum allocated for the land mobile-satellite services and those who wished to preserve the scarce mobile frequency spectrum for maritime and aeronautical users who, unlike land-based users, have no alternative terrestrial communications.[41] The INMARSAT Council, which deferred consideration of this matter until its Twenty-Ninth Session, noted that the outcome of WARC MOB-87 was essentially favorable to INMARSAT.[42] It was highly desirable that INMARSAT should establish an operational land mobile service in the newly-allocated bands at the earliest possible opportunity.[43] It was also noted that there exists a significant demand for integrated position determination and two-way message transfer services, which the Organization should endeavor to satisfy as soon as practicable.[44]

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36. COUNCIL/28/35.
37. *Id. at* sec. 2.
38. Final Acts of the World Administrative Radio Conference for the Mobile Services (Mob-87), Geneva, 1987: the relevant provisions are the partial revisions to article 8 of the Radio Regulations.
39. COUNCIL/28/35, sec. 6.
40. *See supra* note 38.
41. The arguments were reviewed by Olof Lundberg in *Between a Rock and a Hard Place*, TELECOMMUNICATIONS POLICY, March 1987, at 3.
42. COUNCIL/29/SR/FINAL, 19.1.1(a).
43. *Id. at* 19.1.1(b).
44. *Id. at* 19.1.1(c).

The Feasibility Study

At the same Session, in March 1988, the Council considered an initial report by the Director General on the commercial technical, operational and institutional feasibility of providing land mobile-satellite services.[45] The Director General had concluded that a significant market exists for land mobile services and that INMARSAT's future viability would be improved if some of this market could be captured.[46] It would be technically possible to provide a land mobile data service from 1989 using Standard-C.[47] The Director General also considered that a significant potential demand existed for a small, cheap, low-gain telephone terminal. The formulation of concepts for such a terminal, to be known as Standard-M, had already begun.[48]

The Director General presented a revised report to the Thirtieth Session of the Council, held in July 1988. From this the Council noted that there was a significant business opportunity for INMARSAT in the provision of land mobile-satellite services.[49] It decided to authorize the Director General to forward the revised report to the Assembly.[50]

The feasibility report considered the potential market for land mobile services: its findings are set out above.[51] It was noted that INMARSAT would face competition from domestic and regional satellite systems. However, the development of the Standard-C system meant that INMARSAT would be able to meet the needs of land mobile users for data communications and messaging before other competitive systems were in place in most countries. [52] In the future, land mobile voice services could be provided through the development of a small, low-cost voice terminal. Particular emphasis was placed on the importance of achieving worldwide common standards for user terminals.[53]

The report also considered institutional questions. It was noted the INMARSAT constituent instruments allow for considerable flexibility in the provision of communications services. There is no requirement or restriction as to the geographical basis on which INMARSAT's services may be provided: these may therefore be international, regional or domestic. There is no explicit requirement about the classes of users to whom services can be provided, so that services may be offered either to the public or to particular user groups. Furthermore, there is nothing to

45. *Id.* at 15.1.1.
46. *Id. at* 15.1.5(a).
47. *Id. at* 15.1.5(b).
48. *Id. at* 15.1.5(c).
49. COUNCIL/30/SR/FINAL, 14.2.3(a).
50. *Id. at* 14.2.4.
51. *See supra* notes 11, 13, 14 and 16.
52. ASSEMBLY/6/2, ATTACHMENT, sec. 3.3.
53. *Id. at* sec. 4.4.

prevent space segment capacity from being utilized either on a demand-assigned or on a pre-assigned basis.[54] The Convention does require, however, that the space segment shall be open to all users without discrimination on the basis of nationality.[55]

It was also noted that, subject to compliance with international treaty obligations, the use of radiocommunications in national territory is governed by the laws of the State concerned.[56] It should be understood that nothing in the INMARSAT Convention, whether in its original or amended versions, obliges a Member State to permit communications to or from its territory via the INMARSAT space segment. The implications of this are considered below.

The Amendment Process

The procedures for the amendment of the INMARSAT constituent instruments progressed in parallel with the feasibility study. Having circulated the amendments proposed by the Federal Republic of Germany, the Director General received comments from a number of Parties and Signatories, some of whom proposed additional or alternative amendments. The amendments, together with the various comments received, were submitted to the Council for consideration at its Twenty-Ninth Session, in March 1988. The Council already felt able to endorse the principle of amending the Convention and Operating Agreement to enable INMARSAT to provide land mobile-satellite services.[57] It was noted that the Director General would convene a Meeting of Experts appointed by Parties to prepare a consolidated text of the amendments to the Convention, although the Meeting would not consider the proposed amendments to the Operating Agreement. The Party of Bulgaria had issued an invitation for the meeting to be held in Varna, Bulgaria.[58] The consideration of proposed amendments by a Meeting of Experts is not part of the formal procedures for amendment of the Convention. However, such a meeting had proved extremely valuable in achieving consensus on the aeronautical amendments.[59]

The Meeting of Experts, chaired by Mr. B. Barstad of Norway, met in Varna from 20 to 24 June 1988. Thirty parties were represented, and the International Telecommunication Union was represented by an Observer. The Meeting was able to reach a remarkable degree of consensus not only on the substance but also on the text of the proposed amendments for consideration and adoption by the Assembly.[60]

54. *Id.* at sec. 5.2.
55. *Id.* at sec. 5.3.
56. *Id. at* sec. 5.4.
57. COUNCIL/29/SR/FINAL, 15.2.6.
58. *Id.* at 15.2.7(b).
59. *See supra* note 3.
60. ASSEMBLY/6/3, ATTACHMENT, ANNEX V.

The report of the Meeting of Experts was considered by the Council at its Thirtieth Session, in July 1988. The Council decided to recommend to the Assembly the adoption of amendments to the Convention to enable INMARSAT to provide land mobile-satellite services, taking into account the Report of the Meeting of Experts. It also decided to approve the related amendments to the Operating Agreement, and to recommend to the Assembly that it confirm such amendments.[61]

The next regular Session of the Assembly was due to take place in the third quarter of 1989. It was generally agreed that it would be unacceptable for there to be such a delay before the Assembly had the Opportunity to consider the amendments, bearing in mind the time taken for amendments to enter force after adoption by the Assembly. The Council therefore requested the Director General to convene an extraordinary session of the Assembly as soon as possible within the period December 1988 to February 1989 to consider the amendments. The Council also recommended that the Assembly should, as appropriate, reduce the six month interval which is normally required between action on the amendments by the Council and consideration of the amendments by the Assembly.[62]

An extraordinary session of the Assembly was duly convened from 17 to 19 January 1989. This was slightly less than six months after the recommendation of the Council; and the Assembly decided, pursuant to Article 34 of the Convention and Article XVIII of the Operating Agreement, to abridge the interval normally required.[63]

The Assembly adopted the amendments to the Convention and confirmed the amendments to the Operating Agreement with relatively little discussion or disagreement. This reflects great credit on the preparatory work done by the Meeting of Experts, whose consolidated text was adopted with scarcely any modification. On 2 February 1989 the Director General sent a verbal note to all Parties and a letter to all Signatories notifying them of the amendments and reminding them of the final stage in the amendment procedure. Under Article 34(2) of the Convention, the amendments to the Convention will enter into force one hundred and twenty days after the Depositary has received notices of acceptance from two-thirds of those States which at the time of adoption by the Assembly were Parties and represented at least two-thirds of the total investment shares. Under Article XVIII(2) of the Operating Agreement, the amendments to the Operating Agreement will enter into force one hundred and twenty days after the Depositary has received notice of their approval by two-thirds of those Signatories which at the time of confirmation by the Assembly were Signatories and then held at least two-thirds of the total investment shares.

61. COUNCIL/30/SR/FINAL, 4.3.5.

62. *Id.* The relevant periods are prescribed in CONV art. 34(1) and OA art. XVIII(1), which also give the Assembly the power to reduce these periods.

63. *See supra* note 62.